Proprioception-Augmented Risk Mapping for a Commercial Quadruped: Terrain-Aware Navigation Across Legged and Wheeled Configurations

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***Abstract*—Inspection robots build risk maps from geometry, yet slippery or soft ground can look as flat as concrete. This paper studies how proprioceptive evidence should enter a risk map, and whether its value depends on the foot configuration of a commercial quadruped, the DOBOT Rover X1, whose legged and wheeled configurations are fixed before a mission. In MuJoCo, 1400 dynamic episodes on parametrised terrain train a lightweight classifier and measure falls per metre. A separate Monte Carlo study compares geometry-only mapping, local Bayesian updating, appearance-based propagation and an uncertainty-aware fusion on 200 procedural indoor maps with paired statistics. Propagation raised mission survival probability by 0.055 over local updating and fusion added 0.015; both effects held on 200 independent maps, while local updating was practically indistinguishable from geometry. In simulation the legged robot fell on low-friction and soft terrain and the wheeled model never fell, so the benefit of propagation followed the failure model, not perception quality. The classifier reached macro-F1 of 0.935 and 0.999 on held-out episodes but dropped to 0.45 for the legged robot at unseen speeds. All results are simulation only; appearance features remain assumed and no hardware validation is claimed.**

***Index Terms*—legged robots, wheeled-legged robots, proprioception, terrain classification, traversability, risk-aware planning, Monte Carlo simulation**

I. Introduction

Autonomous inspection of underground and industrial spaces depends on knowing which ground can be crossed. Quadrupeds have operated in such settings [1], [2], but geometry says little about friction or softness: a wet floor and dry concrete return the same LiDAR profile, and the robot learns the difference only on contact. Commercial quadrupeds expose joint and inertial signals that could reveal it, behind closed locomotion controllers.

This raises a question separate from the classifier itself: *how should a proprioceptive detection enter the planner's risk map?* It can update only the ground already crossed, or generalise to unvisited ground that looks alike. A second question is whether the answer depends on the feet. The DOBOT Rover X1 uses point-foot and wheel modules that are exchanged by hand (Fig. 1), so the configuration is fixed before a mission; we treat them as two robots and model no switching.

Comparisons of this kind risk circularity: if failure rates are assumed to differ between wheels and legs, finding that the benefit differs restates the assumption. We therefore separate perception from failure and measure both in dynamic simulation where possible. Our contributions are:

1) A reproducible, paired Monte Carlo comparison of four map-update rules, with independent repetition and sensitivity analysis.
2) A dynamic-simulation dataset of both configurations on parametrised terrain, and a lightweight classifier stress-tested beyond its training range.
3) Evidence that the failure model decides how the benefit depends on the feet: simulated falls reverse which configuration gains.

This work follows our inspection architecture on a small servo-driven platform [3], but addresses a different question on a different robot and shares no results with it. We claim no novelty for the fusion rule, a tempered Bayesian update of known form [4].

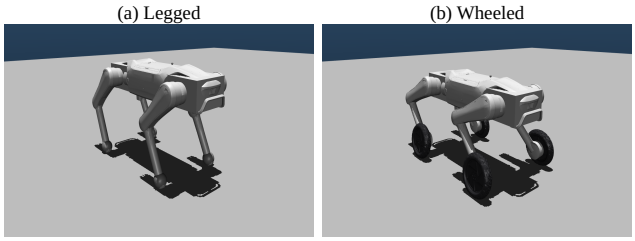


Fig. 1. The two configurations studied, as MuJoCo simulation renders, not photographs: (a) legged, with point feet and 12 actuators; (b) wheeled, with wheel modules and 16 actuators. On the robot the modules are exchanged by hand, so each mission uses one configuration.

II. Related Work

Wheels on a quadruped can greatly reduce the cost of transport [5], and learned controllers achieve robust hybrid navigation [6], but those systems own their controller; here perception and planning act above a closed one. Terrain can be classified from foot forces and joint signals [7], compliance estimated online for control [8], and traversability estimated from legged proprioception [9]. We ask instead how such an estimate should spread through a map. Self-supervised methods project proprioceptive labels onto images [10], Wild Visual Navigation learns traversability online from visual features [11], and terrain-property maps combine semantics with proprioceptive estimates [12]. Those works use real robots and images, which we do not. Our addition is narrower: a controlled, paired measurement

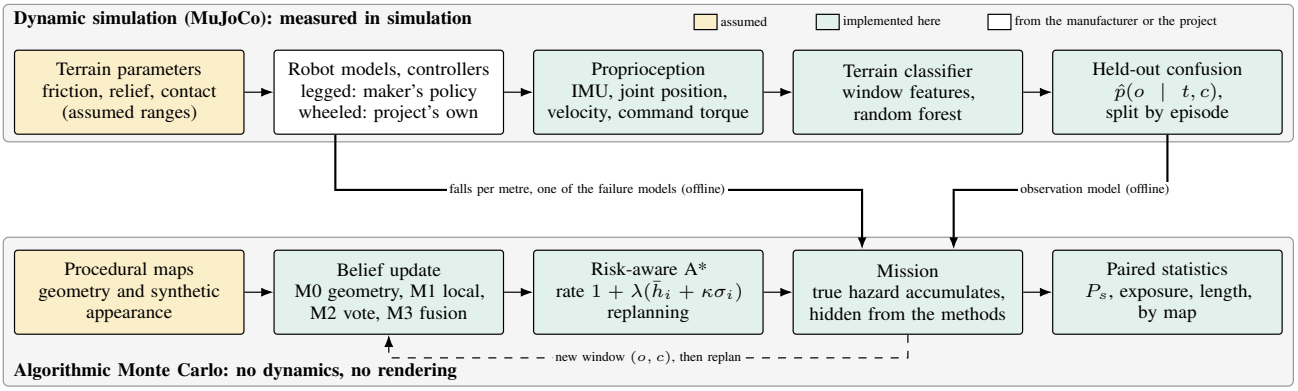


Fig. 2. Study structure. Dynamic simulation supplies the classifier, its held-out confusion and fall rates; these enter the algorithmic Monte Carlo offline, where each method sees only its own observations. Colour gives the status of each block.

of what propagation is worth as classification error and appearance ambiguity vary, across two fixed configurations. Risk-aware planners use CVaR over traversability [13] or evidential uncertainty [14]; our simpler planner is only a common testbed.

III. Platform and Simulation Models

The manufacturer specifies 15.5 kg with point feet and 18.5 kg with wheels, a Livox Mid-360 LiDAR, a RealSense D435i and a Jetson Orin Nano. Dynamics use MuJoCo [15] with models built from the manufacturer's files (the study is outlined in Fig. 2 and its control structure in Fig. 5). The legged model has 12 actuators and walks on the manufacturer's trained policy; the wheeled model has 16 and rolls on a skid-steer controller written for the project, because the manufacturer's is not public. The models weigh 18.65 and 22.08 kg, not rescaled because the source of the difference is unknown. They are not calibrated digital twins.

Validating the models changed the setup. At the project's 200 Hz control rate, the wheeled stance PD (400 N m/rad, 7 N m s/rad) produced a numerical limit cycle of about 70 m/s^2 RMS at the IMU, even on a plane, which a classifier would learn as an artefact. We run physics and the stance PD at 1 kHz, as a joint servo would; mean vertical contact force then matches model weight within 1 % on nominal terrain. The floor has contact priority, since MuJoCo otherwise uses the larger friction of a pair.

IV. Method

Hazard. A mission crosses terrain of true class $t_k \in \mathcal{T} = \{\text{nominal, low friction, granular-like, rough}\}$ over segments ℓ_k. With per-metre hazard $h(t)$ and a patch multiplier m_k, survival is

$$P_s = \exp(-H), \qquad H = \sum_k \ell_k \, m_k \, h(t_k), \tag{1}$$

since independent exponential failure times have additive hazards. $P_s = 0$ if the goal is not reached within three times the shortest path. The hazard is a failure model, distinct from the classifier, and is never derived from F1.

Belief. Each cell i holds $b_i(t)$. Geometry observes relief only, so the prior separates rough terrain and nothing else:

$$b_i^0(t) \propto \pi(t)\, \mathcal{N}(r_i; \mu_t, \sigma_g^2). \tag{2}$$

Every travelled window of ℓ_w metres yields a classifier output o and confidence bin c, with likelihood $\hat{p}(o \mid t, c)$ estimated on calibration episodes. It is the add-half estimate

$$\hat{p}(o \mid t, c) = \frac{n_{t,o,c} + \frac{1}{2}}{\sum_{o'} n_{t,o',c} + \frac{1}{2}|\mathcal{T}|}, \tag{3}$$

with $n_{t,o,c}$ the count of calibration windows of class t, output o and confidence bin c (maximum class probability below or above 0.8). Windows are assumed conditionally independent given t, which is optimistic and motivates the tempering below.

Update rules. **M0** keeps (2). **M1** applies Bayes' rule only to the cells of the traversed window:

$$b_i(t) \leftarrow b_i(t)\, \hat{p}(o \mid t, c) \Big/ \sum\nolimits_{t'} b_i(t')\, \hat{p}(o \mid t', c). \tag{4}$$

M2 is an online appearance vote. With kernel $s_i^w = \exp(-\|a_i - \bar{a}_w\|^2 / 2\sigma_k^2)$ between the appearance feature of cell i and the mean feature of window w, votes are $V_i(k) = \sum_w s_i^w \mathbf{1}[o_w = k]$. Where $\sum_k V_i(k) \geq \tau$, the belief is replaced by the winning label, taken as correct. **M3** instead applies a tempered update,

$$\begin{aligned} \log b_i(t) = {} & \log b_i^0(t) + D_i(t) \\ & + \max\big[P_i(t) - \max_{t'} P_i(t'),\, -C\big] + \text{const}, \end{aligned} \tag{5}$$

where D_i sums the log-likelihoods of windows that crossed cell i and $P_i(t) = \rho \sum_w s_i^w \log \hat{p}(o_w \mid t, c_w)$. The exponent $\rho < 1$ discounts correlated evidence, as in power posteriors [4]; C keeps propagated evidence from reaching certainty.

Appearance. Each cell carries a feature $a_i = p_{v(i)} + \delta_{j(i)} + \epsilon_i$, with p_v a per-map prototype of its visual type, δ_j a patch offset and ϵ_i cell noise. The terrain-to-look relation is random per map and partly aliased, so it can only be exploited online. It stands in for a visual embedding.

Planning. With believed hazards $\hat{h}(t)$, the expected rate is $\bar{h}_i = \sum_t b_i(t) \hat{h}(t)$. By Jensen's inequality $-\log \mathbb{E}[e^{-H}] \leq$

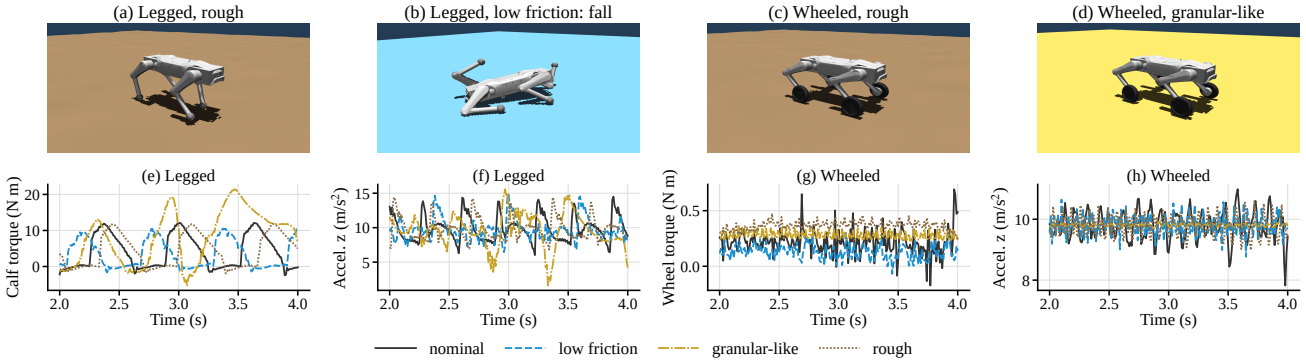


Fig. 3. MuJoCo simulation. (a)–(d) Renders of campaign episodes, with floor colour added for display only. (e)–(h) Logged signals over 2 s on the four terrains, from example episodes that lasted at least 5.5 s. Commanded torque is the actuator command, not a contact force.

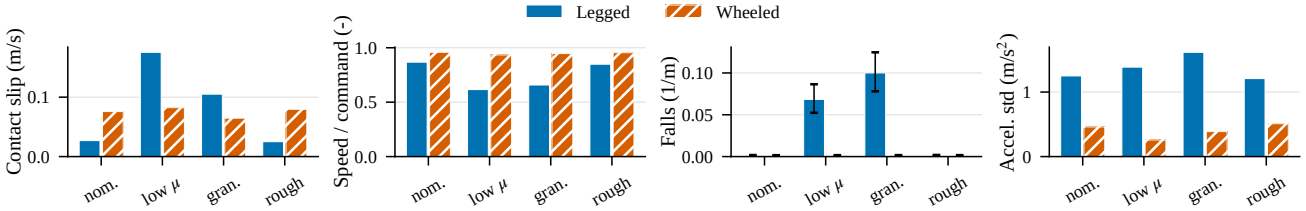


Fig. 4. Dynamic behaviour by configuration and terrain, with 95 % credible intervals on falls. Wheeled slip and acceleration come from a different controller and contact geometry, and are not directly comparable with legged values.

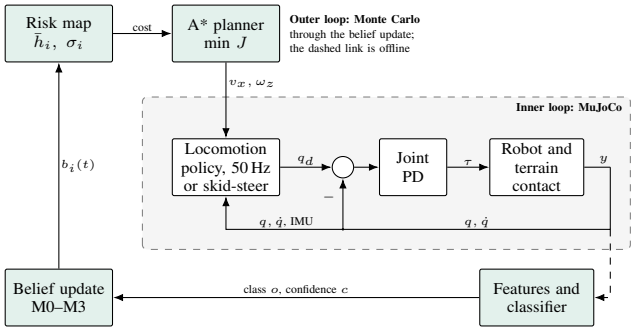


Fig. 5. Control structure. The inner loop runs in MuJoCo; y is the proprioceptive output. The outer loop runs in the Monte Carlo and is joined to the inner loop offline (dashed), through the classifier's confusion. The two were not closed online in one simulator.

$\mathbb{E}[H]$, so $\sum \ell\,\bar{h}$ upper-bounds the negative log survival under the belief. A* minimises

$$J = \sum_k \ell_k \big[1 + \lambda\big(\bar{h}_{i_k} + \kappa\,\sigma_{i_k}\big)\big], \tag{6}$$

with σ_i the posterior standard deviation of the hazard ($\kappa = 0$ except in M3). Costs are at least 1 per metre, so the octile distance is admissible. Replanning follows each window; an update costs $O(N|\mathcal{T}|)$ and a search $O(N \log N)$ for N cells.

V. Experimental Setup

Dynamic simulation. Each configuration ran 150 episodes per class, plus 50 on ramps and 50 on mixed courses (1400 in total), with a fresh terrain instance and a command of 0.4–0.9 m/s and 0 or ±0.35 rad/s. Table I lists the assumed ranges. Granular-like means soft, lossy contact, not particles, and MuJoCo friction is a parametrised model, not

TABLE I
Terrain Parameters per Class (Assumed Ranges)

Terrain	Friction μ	Relief (mm)	Contact time constant (s)
Nominal	0.70–1.20	0–6	0.02
Low friction	0.12–0.30	0–6	0.02
Granular-like	0.45–0.80	5–15	0.05–0.15
Rough	0.70–1.20	25–60	0.02
Ramp (5–12°, test only)	0.70–1.20	none	0.02

a measurement of real floors. A fall is a body tilt beyond 30°. Ramps and mixed courses never enter training; each mixed course crosses all four terrains in random order, 5 m each, so that windows straddle boundaries.

Classifier. Features come from 2 s windows at 100 Hz: IMU (acceleration, angular rate, tilt) and joints (position, velocity, commanded torque, not contact force), plus the command. Each channel gives its standard deviation, first-difference RMS and log band powers in 0.5–5, 5–15 and 15–45 Hz, for 34 IMU features and 48 (legged) or 60 (wheeled) joint features. We add assumed noise and bias, and drop the last 1 s before a fall. Contact forces, true velocity and slip are logged for validation only and never enter the classifier. We use a random forest and a logistic regression, with five-fold cross-validation *by episode* and a 60/20/20 train/calibration/test split for the observation model.

Monte Carlo. Maps have 48×32 cells of 0.5 m with drifts, crossings, rooms and pillars, and about 31 % hazard coverage. Appearance is synthetic: per-map random prototypes with noise, so no terrain-to-look relation carries between maps; 15 % of hazard patches look like floor, and decoys make safe ground look hazardous. The Monte Carlo has no rendering, SLAM error or dynamics, and

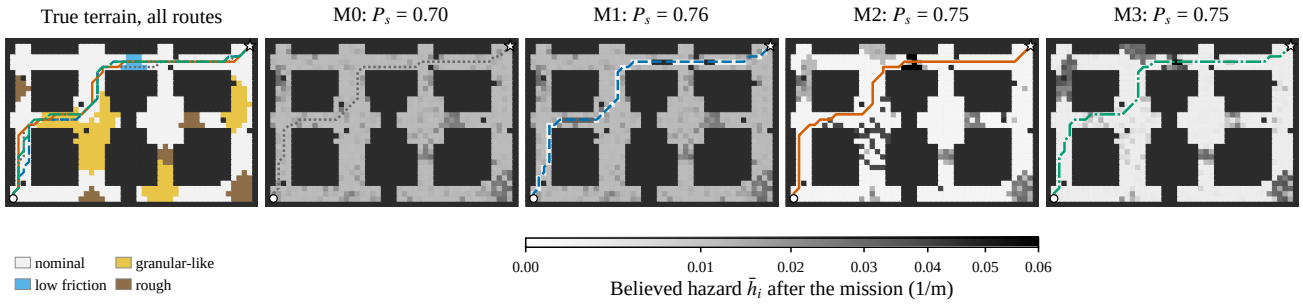


Fig. 6. One evaluation map (id 12, fixed in advance) at $\varepsilon = 0.1$: true terrain with the four routes, and the hazard each method believes at mission end. Dots: start; stars: goal; dark cells: rock or high believed hazard. Here M1 happens to beat M2 and M3: the mean gain does not hold on every map.

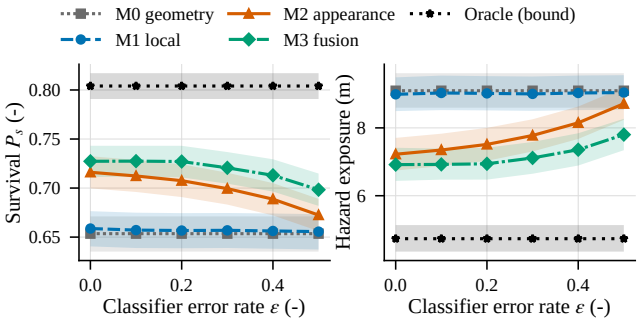


Fig. 7. Robustness to classifier error (mean and 95 % interval, 200 maps; synthetic symmetric confusion).

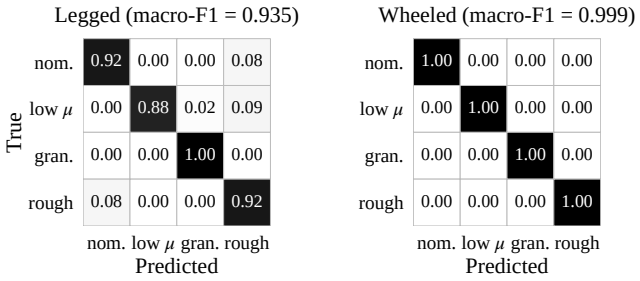


Fig. 8. Confusion on held-out episodes (fusion, forest), normalised by true class; source of the measured observation model.

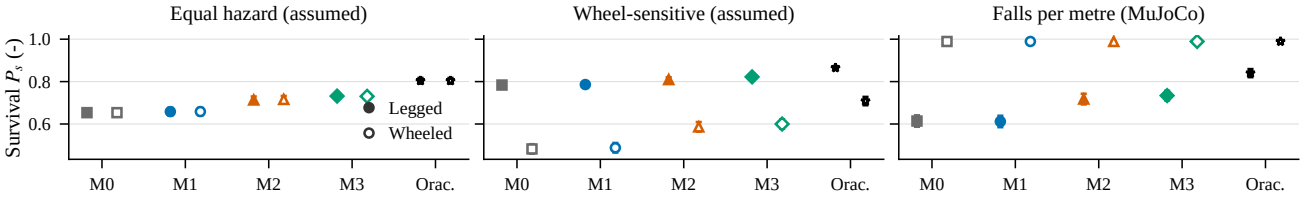


Fig. 9. P_s with measured classifier confusion (Fig. 8) under three failure models (mean and 95 % interval, 200 maps). Filled: legged; open: wheeled. The configuration that gains most depends on the failure model.

TABLE II
CLASSIFIER BY EPISODE: MACRO-F1 [95 % INTERVAL] (BRIER)

Macro-F1 (Brier score)	Legged	Wheeled
IMU, forest	0.939 [0.926, 0.951] (0.134)	0.997 [0.995, 0.998] (0.015)
Joints, forest	0.911 [0.895, 0.926] (0.178)	0.990 [0.987, 0.993] (0.022)
IMU + joints, forest	0.956 [0.943, 0.966] (0.102)	0.998 [0.997, 0.999] (0.010)
IMU + joints, logistic	0.963 [0.955, 0.970] (0.074)	0.998 [0.997, 0.999] (0.004)
Stress tests, IMU + joints, forest		
Random window split (leaky)	0.982	0.999
Held-out 20 % of episodes	0.935	0.999
Sensor noise ×3	0.951	0.997
Mixed terrain course	0.820	0.842
Unseen speed range (worse of two)	0.450	0.963
Unseen terrain parameters (worse of two)	0.696	0.935

TABLE III
MONTE CARLO AT $\varepsilon = 0.1$, EQUAL ASSUMED HAZARD (200 MAPS, 95 % BOOTSTRAP INTERVALS)

Method	P_s	Success	Exposure (m)	Path (m)	Replans
M0 geometry	0.654 [0.635, 0.671]	0.676	9.10 [8.59, 9.61]	32.6	0.0
M1 local	0.657 [0.639, 0.675]	0.678	9.03 [8.52, 9.54]	32.6	4.5
M2 appearance	0.712 [0.696, 0.729]	0.728	7.35 [6.87, 7.83]	33.0	19.5
M3 fusion	0.727 [0.712, 0.743]	0.750	6.92 [6.47, 7.39]	33.0	12.1
Oracle (bound)	0.804 [0.791, 0.817]	0.824	4.73 [4.35, 5.14]	33.4	0.0

obstacles are known. A mission fails analytically through (1); realised success draws one exponential threshold per

TABLE IV
PAIRED DIFFERENCES IN P_s [95 % BOOTSTRAP INTERVAL]

Condition	M1 − M0	M2 − M1	M3 − M2
Synthetic ε=0.1	+0.004 [+0.000, +0.007]	+0.055 [+0.043, +0.067]	+0.015 [+0.008, +0.023]
Synthetic ε=0.3	+0.003 [+0.001, +0.006]	+0.043 [+0.031, +0.054]	+0.021 [+0.015, +0.028]
Repeat, ε=0.1	+0.001 [−0.002, +0.004]	+0.049 [+0.038, +0.059]	+0.014 [+0.008, +0.019]
Held-out conf., legged	+0.005 [+0.002, +0.008]	+0.055 [+0.043, +0.067]	+0.018 [+0.010, +0.027]
Held-out conf., wheeled	+0.005 [+0.002, +0.009]	+0.057 [+0.045, +0.070]	+0.015 [+0.005, +0.024]
Course conf., legged	+0.005 [+0.002, +0.008]	+0.044 [+0.033, +0.055]	+0.027 [+0.019, +0.036]
Course conf., wheeled	+0.004 [+0.001, +0.007]	+0.047 [+0.036, +0.059]	+0.023 [+0.015, +0.031]
MuJoCo falls, legged	−0.002 [−0.007, +0.002]	+0.106 [+0.086, +0.127]	+0.016 [+0.003, +0.030]
MuJoCo falls, wheeled	+0.000 [−0.000, +0.000]	+0.000 [+0.000, +0.000]	+0.000 [−0.000, +0.000]

replicate, shared across methods, so that all methods face common random numbers. Evaluation uses 200 maps with five replicates and common random numbers; the repetition uses 200 unseen maps. Parameters were tuned only on 50 development maps (σ_k=0.35 for M2 and 0.5 for M3, τ=0.5, ρ=0.6, κ=1; the last three at the edge of their grids). Fixed assumptions are $\lambda = 25$, $\ell_w = 1\,\mathrm{m}$ and hazard $[0.001, 0.06, 0.04, 0.03]\,\mathrm{m}^{-1}$ over $\mathcal{T}$, swept in Section VI. In all, 177 000 evaluation missions were run.

Statistics. The map is the unit; replicates are averaged within it. We report 95 % bootstrap intervals of paired differences (10 000 map resamples), Wilcoxon tests with

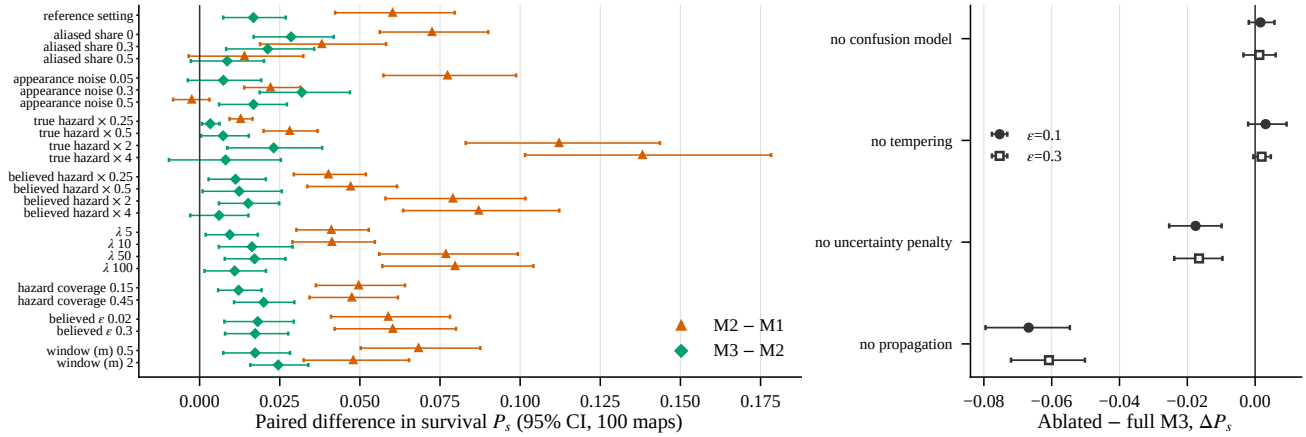


Fig. 10. Left: sensitivity of the paired gains to assumed parameters at $\varepsilon = 0.1$. Right: ablation of M3 on 200 maps. Intervals crossing zero show no measurable effect.

Holm correction, and d_z. Hypotheses and refutation criteria were recorded before the campaigns: **H1** M2 beats M1 and M0; **H2** M1 beats M0; **H3** M3 beats M2; **H4** IMU plus joints beats each alone; **H5** the propagation gain differs between configurations through measured perception alone.

VI. Results and Discussion

Physical behaviour. The legged robot fell in 61/150 low-friction and 70/150 granular-like episodes and never on nominal or rough terrain, giving Jeffreys-posterior hazards of 0.068 and 0.100 m^{-1}, while contact slip rose from 0.027 to 0.175 m/s on low friction. The wheeled model never fell in 650 episodes, including ramps. The wheeled controller is ours, with a stiff stance, and has no fidelity claim, so the result describes these controllers only (Figs. 3 and 4).

Classification. On 495 legged and 600 wheeled episodes (Table II), a random window split inflates legged macro-F1 from 0.956 to 0.982, which measures the leakage avoided. Held-out recall per class is 0.92, 0.88, 1.00 and 0.92 for nominal, low friction, granular-like and rough on the legged robot. Logistic regression matches or exceeds the forest on the legged robot (0.963 against 0.956) with lower calibration error (0.023 against 0.107); the forest stayed as the observation model, as fixed in advance. Tripling the assumed sensor noise changes little (0.951 and 0.997). H4 holds for the legged robot: fusion beats IMU by [+0.0100, +0.0248] and joints by [+0.033, +0.057]. For the wheeled robot the interval against IMU reaches zero, so H4 is not established there. The high scores say little about real ground: classes are separated by construction, and a wheeled F1 of 0.998 mostly shows how clean simulated signals are. The stress tests are more informative. Mixed courses give 0.820 and 0.842; with the legged robot, unseen speeds give as little as 0.450 (0.658 with logistic regression) and unseen terrain parameters 0.696. Ramps, never trained on, are read as a hazard class in 99 % (legged) and 79 % (wheeled) of windows. The classifier interpolates within its training range and should not be trusted beyond it. Inference takes about 10 ms on a laptop core, not measured on the Jetson.

Map-update rules (Tables III and IV, Figs. 6 and 7). **H1 is supported**: at $\varepsilon = 0.1$, M2 improves P_s over M1 by +0.055 [+0.043, +0.067], d_z=0.64, and the repetition gives +0.049 [+0.038, +0.059]. Against M0, M2 and M3 cut the accumulated hazard by 21 % and 26 %. The cost is 0.37 m of path and more replanning (19.5 against 4.5 replans); a search took 3.3–4.0 ms on average in Python on a laptop. On one replicate per map, realised success was 0.665 for M0 (Wilson [0.597, 0.727]) and 0.735 for M3 ([0.670, 0.791]). **H2 is not established**: M1 improves on M0 by only +0.004 [+0.000, +0.007], Holm-adjusted p=0.055, and the repetition interval [−0.002, +0.004] includes zero. Local updates arrive after the ground is crossed. A stronger baseline, added after internal review, also spreads evidence to neighbouring cells by distance alone (reach of 2 m, tuned on development maps): it gains only +0.002 [−0.002, +0.006] over M1, and M2 still beats it by +0.053 [+0.042, +0.065], so appearance, not mere spatial spread, carries the gain. **H3 is supported with a small effect**: M3 adds +0.015 [+0.008, +0.023] over M2, +0.014 on repetition, growing with classifier error (+0.026 at $\varepsilon = 0.5$). All methods stay well below the oracle bound of 0.804.

Does the benefit depend on the feet? With equal assumed hazards and only the measured confusion differing, the M2−M1 gain is +0.055 (legged) and +0.057 (wheeled). Their difference, +0.002 [+0.001, +0.004], excludes zero, so H5 survives its recorded criterion, but the effect is negligible: perception quality creates no meaningful difference between configurations. Fig. 9 shows what does. With an assumed table that penalises wheels, the gain is larger for wheels (+0.098 against +0.023). With simulated falls per metre (Fig. 4), the ordering reverses: legged +0.106, wheeled +0.000, because wheeled P_s is already 0.989 under M0, a level set by the Jeffreys prior since no fall occurred. The Monte Carlo returns whichever failure model it is given, so an unmeasured table should not be used to claim that one configuration benefits more. Mixed-course confusion lowers the M2 gain while M3 holds (Table IV).

Sensitivity and ablation. The propagation gain depends

on the assumed appearance channel (Fig. 10): it falls from +0.072 with no aliasing to +0.014 [−0.003, +0.032] when half the patches are aliased, and to −0.002 [−0.008, +0.003] at high appearance noise. It scales with hazard, from +0.013 to +0.138 across a 16-fold range. A longer window lowers the gain (+0.068 at 0.5 m, +0.048 at 2 m), and a wrong belief about classifier error barely moves M3 over M2 (+0.018 and +0.017 when $\varepsilon = 0.1$ is believed to be 0.02 or 0.3). In M3, removing the confusion model (+0.002 [−0.002, +0.006]) or the tempering (+0.003 [−0.002, +0.009]) has no measurable cost, so these are not supported as contributors. Removing propagation costs −0.067 and the uncertainty penalty −0.018. A higher λ partly replaces that penalty: at $\varepsilon = 0.3$, $\lambda = 400$, M3 without it reaches 0.744, M3 0.743 and M2 0.728. Even with each method at its best swept λ, M3 exceeds M2 by +0.013 [+0.005, +0.021] at $\varepsilon = 0.1$, so the soft fusion appears to account for the persistent part. This sweep is descriptive.

Implications. Three points follow, conditional on this simulation. First, a local update alone arrives too late to change the route; its value is as a source of labels for something that can generalise. Second, propagation is only as good as the appearance channel, so the first measurement to make is how well the robot's cameras separate physically different terrains, including visually similar pairs. Third, failure must be measured for each configuration with its real controller before claiming that one benefits more; our fall rates may well change on the robot.

Limitations. No result comes from the physical robot, and none validates a real mine. The Monte Carlo is algorithmic: appearance is synthetic, so H1 rests on an assumption that vision must earn, and work on real images [10], [11] is the better guide to it. Main-study hazards are assumed. Falls are a narrow notion of failure, from short episodes on homogeneous ground. Joint signals have not been captured from the robot, and the sim-to-real gap of the classifier is unknown and likely large. Gazebo produced no result here: it follows the pose computed by MuJoCo. Under ROS 2 Jazzy its legged motion and sensor gates passed; the wheeled motion gate failed (0.184 m/s after release).

VII. Conclusion

In this simulation study, generalising proprioceptive evidence by appearance raised mission survival, while updating only the crossed ground did almost nothing. A soft, uncertainty-penalised fusion added a small gain that grew with classifier error. The failure model, not perception, decided whether the benefit favours legs or wheels. Next steps are capturing joint signals on the robot under supervision, replacing synthetic appearance with camera features, and measuring failure on real surfaces.

Reproducibility

The simulator, capture and classifier code, per-mission records with their seeds, evaluation maps, window features and the scripts that regenerate every number, table and figure are available at https://github.com/JoseSegundoM/proprioceptive_risk_mapping. The robot models come from the project repository `mine_sentinel_autonomy` at commit `eca13b0`, and raw MuJoCo signals are regenerated from their recorded seeds.

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